SILENT SOS

Emergency Alert Website

PROJECT REPORT
A web-based platform for emergency contact management, demo SOS alerts, location-sharing information and alert monitoring.

Prepared by

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Project interface screenshots are included in this report.

1. Project Overview

Silent SOS is an emergency alert management website designed to help users organise trusted emergency contacts and view emergency-related information from a single interface. The website includes a Home page, an Emergency Contacts page and a Dashboard. The interface also presents an SOS demo alert, location-sharing information and SOS alert history.

The Home page states that the alert experience is in demo mode and that real SMS or WhatsApp alerts will not be sent. This report describes the visible interface and intended workflow shown in the supplied screenshots.

2. Problem Statement

During an emergency, users may need quick access to trusted contact details and a clear way to review emergency alert information. Silent SOS brings these related tasks together in one website interface.

3. Project Objectives

- Provide a simple and easy-to-understand emergency website interface.
- Allow users to enter and manage trusted emergency contact details.
- Provide an SOS demo alert interface.
- Present location-sharing information and a View Location action.
- Display alert statistics and alert history in a dashboard.
- Make active and resolved alert states easy to distinguish.

4. Main Modules

Module	Purpose
Home Page	Introduces Silent SOS, provides navigation, an SOS demo button and key feature cards.
Emergency Contacts	Provides fields for full name, phone number, optional email and relationship, plus a Save Contact action.
Dashboard	Shows summary cards, saved emergency contacts, shared locations and SOS alert history.

5. Home Page

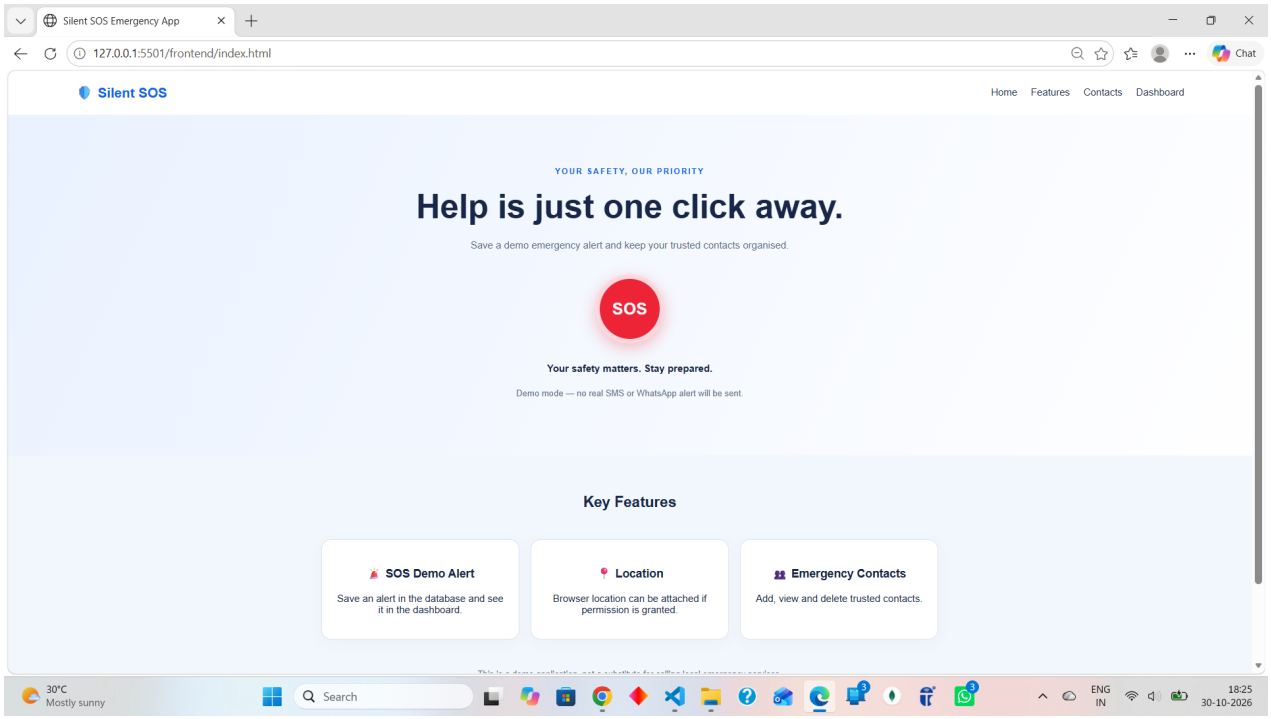


Figure 1. Silent SOS Home page showing navigation, the SOS demo button and Key Features.

The Home page is the starting point of the website. The navigation bar provides access to Home, Features, Contacts and Dashboard. The central heading, “Help is just one click away,” communicates the purpose of the application. A prominent red SOS button represents the demo emergency-alert action. The Key Features area highlights SOS Demo Alert, Location and Emergency Contacts.

The page explicitly displays a demo-mode notice: real SMS or WhatsApp alerts will not be sent. This distinction is important when presenting the project.

6. Emergency Contacts Page

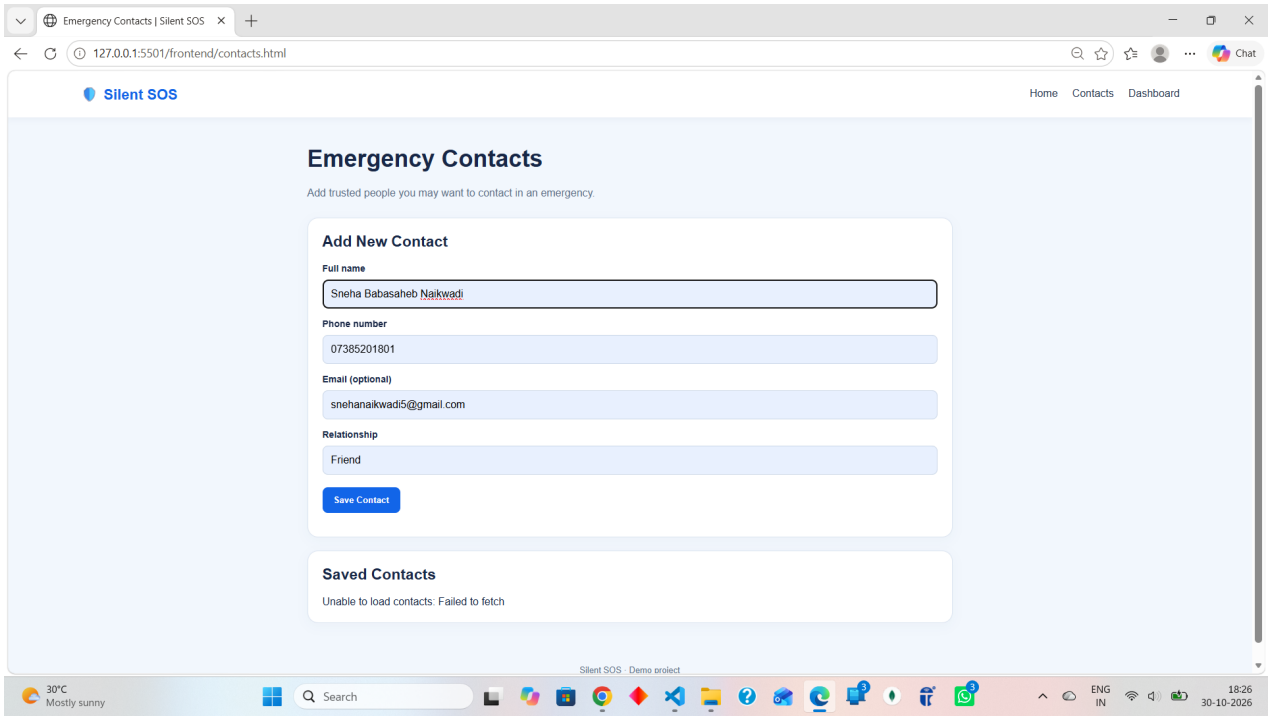


Figure 2. Emergency Contacts page showing the Add New Contact form and Saved Contacts section.

The Emergency Contacts page is used to enter trusted contact information. The form contains Full name, Phone number, Email (optional) and Relationship fields. The Save Contact button submits the form. The Saved Contacts section is intended to display the contacts added by the user.

In the supplied screenshot, the contact form is populated with sample details, but the Saved Contacts section displays “Unable to load contacts: Failed to fetch.” This indicates that the screenshot does not demonstrate a successful live retrieval from the backend. The report describes the feature's purpose without claiming that the data was successfully saved or loaded in this capture.

7. Dashboard

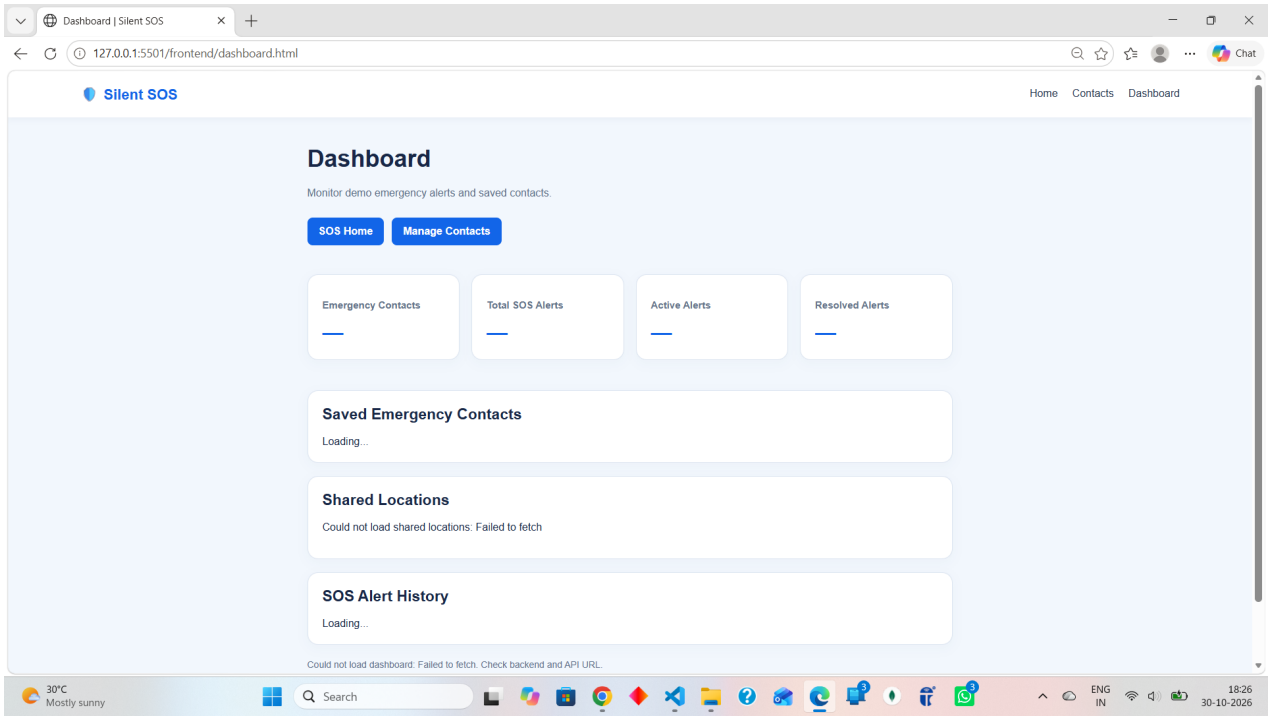


Figure 3. Dashboard page showing statistic cards, Saved Emergency Contacts, Shared Locations and SOS Alert History.

The Dashboard brings together emergency-related information. At the top, the page provides SOS Home and Manage Contacts navigation buttons. Four statistic cards are provided: Emergency Contacts, Total SOS Alerts, Active Alerts and Resolved Alerts.

The Saved Emergency Contacts panel is intended to show saved contact records. The Shared Locations panel is intended to show location-sharing details and a View Location action. The SOS Alert History panel is intended to show past alerts and their statuses, including active and resolved alerts.

In the supplied screenshot, the statistic cards show placeholder dashes, the contacts and alert-history panels show Loading, and the shared-locations panel shows “Failed to fetch.” A message at the bottom asks the user to check the backend and API URL. This suggests that the frontend could not retrieve the required data at the time the screenshot was taken.

8. Website Workflow

- The user opens the Home page and reviews the available features.
- The user navigates to Emergency Contacts.
- The user enters a contact's name, phone number, optional email and relationship.
- The user selects Save Contact.
- The user opens the Dashboard to review summary cards and emergency information.
- The user can review the Shared Locations section and use the View Location action when location data is available.
- The user can review SOS Alert History and the status of available alerts.

9. Technologies and Architecture

The project implementation discussed for this website uses HTML, CSS and JavaScript on the frontend. The backend is built with Node.js and Express.js, and MongoDB is used as the database. The frontend communicates with backend API endpoints to request or submit information. The supplied screenshots show the frontend interface, but do not independently verify that the backend connection is working.

- HTML: Defines the structure of pages, forms, buttons and content sections.
- CSS: Controls layout, spacing, colours, typography and responsive appearance.
- JavaScript: Handles page interactions, form actions and API requests.
- Node.js and Express.js: Provide the server and API routes.
- MongoDB: Stores application records such as contacts and alerts.

10. Current Observations and Limitations

The supplied screenshots confirm the visual structure of the three pages. They also show that the Contacts and Dashboard pages were unable to fetch some backend data at capture time. Before presenting the website as fully connected, the backend service, API base URL, database connection and browser network requests should be checked.

The Home page identifies the SOS feature as demo mode. Therefore, real SMS or WhatsApp delivery should not be claimed based on these screenshots. Location sharing should be described as permission-based and should only be presented as live when an actual consented location record is available.

11. Future Enhancements

- Complete and verify the backend API connection for contacts and dashboard data.
- Add real-time SMS or WhatsApp notifications after configuring an approved messaging service.
- Enable consent-based GPS location sharing and map display.
- Add user authentication and role-based access.
- Improve error handling and show clear messages when the backend is unavailable.
- Test the website on mobile devices and different browsers.

12. Conclusion

Silent SOS brings emergency contact management, an SOS demo interface, location-sharing information and alert monitoring together in one website. The three main pages provide a clear structure: the Home page introduces the system, the Emergency Contacts page collects trusted contact details, and the Dashboard presents emergency information in one place. The screenshots also identify backend-fetch issues that should be resolved and tested before the system is described as fully operational.

13. Acknowledgement

I would like to thank my teachers, guides and everyone who supported me during the development of this project. This project helped me understand website design, frontend and backend communication, API

integration and database concepts.